

THOR: A Versatile Foundation Model for Earth Observation Climate and Society Applications

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Abstract

Current Earth observation foundation models are architecturally rigid, struggle with heterogeneous sensors and are constrained to fixed patch sizes. This limits their deployment in real-world scenarios requiring flexible compute-accuracy trade-offs. We propose THOR, a "compute-adaptive" foundation model that solves both input heterogeneity and deployment rigidity. THOR is the first architecture to unify data from Copernicus Sentinel-1, -2, and -3 (OLCI & SLSTR) satellites, processing their native 10 m to 1000 m resolutions in a single model. We pre-train THOR with a novel randomized patch and input image size strategy. This allows a single set of pre-trained weights to be deployed at inference with any patch size, enabling a dynamic trade-off between computational cost and feature resolution without retraining. We pre-train THOR on THOR Pretrain, a new, large-scale multi-sensor dataset and demonstrate state-of-the-art performance on downstream benchmarks, particularly in data-limited regimes like the PANGAEA 10% split, validating that THOR's flexible feature generation excels for diverse climate and society applications. Code is available at <https://github.com/FM4CS/THOR>.

1. Introduction

Earth Observation (EO) enables large-scale monitoring of Earth's systems (e.g., [9, 11, 19, 24]), but presents a monumental computer vision challenge. Foundation models (FM) promise to solve EO [17], but applying models pre-trained on natural images is often sub-optimal [25] because one must ingest vast, heterogeneous data streams from diverse sensors (e.g., optical, SAR) at scales from meters to kilometers ground sampling distance (GSD).

Most EO-specific FMs (e.g., [8, 13, 26, 30, 32]), often built on Vision Transformers (ViT), are architecturally rigid. They are trained using fixed input image and patch sizes (e.g., 16×16), which create a bottleneck for data-efficient adaptation: coarse patching produces low-resolution to-

ken sequences. Consequently, dense tasks like segmentation require large, complex decoders (e.g., UPerNet [31]) to upsample features. These decoders demand significant amounts of labeled data for fine-tuning, undermining the data-efficiency promise of FMs.

To address these shortcomings, we propose THOR (Transformer based foundation model for Heterogeneous Observation and Resolution), a versatile multi-modal FM. THOR is the first architecture to both unify the 10 m - 1000 m GSD range of Sentinel-1, -2, and -3 (including the SLSTR sensor) and integrate a compute-adaptive patching strategy, solving both input heterogeneity and deployment rigidity. By incorporating a randomized patch size and input image size during pre-training, THOR becomes "compute-adaptive", allowing a single set of weights to be deployed with various configurations. Users can select smaller patch sizes to produce denser, higher-resolution token sequences for tasks like fine-grained classification and use lightweight decoders. Conversely, lower token density reduces memory and compute requirements, suiting global tasks like climate trend analysis and ocean monitoring, or scenarios with sufficient training data to support larger, more complex decoders. Moreover, multi-sensor integration allows THOR to leverage synergies: radar from Sentinel-1, multi-spectral details from Sentinel-2, and broad-scale climate context from Sentinel-3 within a single model. Sentinel-3 provides daily global coverage, essential for monitoring high-frequency phenomena like snow melt and ocean color.

To learn the model, we created THOR Pretrain dataset, a new, large-scale dataset of 22TB that has been aligned spatio-temporally and across modalities. It is the first to unify Sentinel-1, -2, and -3 (both OLCI and SLSTR) data, at native resolutions from 10 m to 1000 m. THOR Pretrain also contains land cover products, digital elevation models (DEM), and ERA5-Land variables.

In summary, our key contributions are as follows:

- A flexible, multi-sensor architecture that is the first FM to unify Sentinel-1 SAR, Sentinel-2 MSI, and Sentinel-3 OLCI & SLSTR data from 10 m - 1000 m GSD, built on a compute-adaptive backbone (Sec. 4.1).

- A novel multi-modal pre-training framework that extends the flexible patching to a Masked Autoencoder (MAE) setup, combining pixel-level reconstruction with pretext tasks for land cover and climate variables (Sec. 4.3).
- THOR Pretrain: A new, large-scale and diverse multi-modal EO dataset, curated with a novel sampling strategy to ensure geographic and thematic diversity (Sec. 3).

We demonstrate that this co-design achieves state-of-the-art performance in limited training data regimes (Sec. 5).

2. Related work

Self-supervised pre-training strategies in EO. Recent EO FMs leverage self-supervised learning (SSL) [28], primarily MAE (e.g., Prithvi-EO-2.0 [26], MMEarth [22], SatMAE [5]) and hybrid contrastive methods (e.g., CROMA [8]). While powerful, these models are architecturally rigid. Prithvi-EO-2.0 is pre-trained exclusively on 30 m GSD data [26], MMEarth adopts a "resample-to-grid" strategy, harmonizing data to a 10 m grid and discarding resolution information [22], and CROMA adopts a contrastive objective for radar-optical invariance and MAE reconstruction [8]. All use fixed patch sizes (e.g., 16×16 or 8×8). This locks in a specific computational profile and, as argued in the introduction, necessitates large, data-hungry decoders for dense tasks.

Architectural solutions for input heterogeneity. State-of-the-art models like TerraMind [13], DOFA [32], and Copernicus-FM [30] employ sophisticated input data processing strategies, such as TerraMind's "dual-scale early fusion" for nine modalities or DOFA's wavelength dependent "dynamic weight generator" serving as a flexible translation layer for heterogeneous sensor data. AnySat [3] achieves this via scale-adaptive spatial encoders and introducing Joint Embedding Predictive Architecture [2] for multi-modal EO data and leverages the spatial alignment of multiple modalities for self-supervision. Copernicus-FM unifies Sentinel missions (Sentinel-1, -2, -3 OLCI, and -5P), covering 10 - 1000 m GSD [30]. An "extended dynamic hypernetwork" generates metadata-based weights to make it an "input-flexible" model, but it lacks deployment versatility. Scale-MAE [23] makes positional encoding "scale-aware" by scaling it by the image's GSD. However, it only handles one modality at a time, and its fixed patch size leads to inconsistent sequence lengths and computational loads for the same ground area. USat [12] uses separate patch projection layers for different bands. While this effectively ingests multi-modal data, USat's architecture remains rigid. Its "superpositional encoding" scheme is structurally constrained and incompatible with a flexible patching strategy required for compute-adaptive inference. While these models handle diverse inputs, they are "deployment-rigid", using a fixed patch and input image size during pre-training. For instance, Copernicus-FM is trained with a fixed foot-

print, limiting Sentinel-5P images to a few pixels.

Architectural rigidity and adaptive models. FlexiViT [4] demonstrated that randomizing patch sizes during pre-training, a single set of ViT weights can perform compute-adaptive inference with variable patch sizes. EO FMs are adopting this concept. Galileo [27] incorporates resizable patch embeddings from FlexiViT, and pairs it with a dual-objective training strategy to capture both high-level semantic global context and fine-grained local details. Similarly, FlexiMo [16] utilizes the FlexiViT strategy and includes a "wavelength-guided channel adaptation" module to handle multi-sensor inputs, allowing the pre-trained model to adapt to arbitrary spatial resolutions and maintain multi-scale feature fidelity. While these models are a key step towards deployment versatility, they focus on Sentinel-1 and -2, without scaling to the full multi-resolution challenge (10 m - 1000 m) posed by sensors like Sentinel-3 OLCI & SLSTR.

The gap: synthesizing input and deployment. The related work reveals two powerful, yet until now, parallel lines of research. On one side, models like Copernicus-FM and USat solve input heterogeneity but are deployment-rigid. On the other, models like Galileo and FlexiMo solve deployment versatility but have not been scaled to the full multi-resolution (10 m - 1000 m) challenge. A critical gap therefore exists: no single architecture has solved both input heterogeneity and deployment versatility simultaneously. Our work, THOR, is designed to be the first to fill this gap. This challenge is non-trivial, as it requires co-designing the positional encoding, per-band patch projection, and MAE loss function to be mutually compatible across a 100x GSD range (10 m to 1000 m). We propose a new architecture that synthesizes state-of-the-art approaches for multi-sensor input with compute-adaptive patching, enabling a single model to operate efficiently across the full 10 m - 1000 m GSD range. We detail this architecture in Sec. 4.

3. THOR Pretrain

THOR is pre-trained on a new, diverse, and large-scale dataset named THOR Pretrain. This dataset is curated to learn representations that are robust to variations in global land cover, ocean phenomena, and cloud conditions.

THOR Pretrain unifies data from four major Copernicus Sentinel sensors: Sentinel-1 SAR, Sentinel-2 MSI, Sentinel-3 OLCI and SLSTR. These sensors provide diverse image modalities, including radar, multispectral and thermal, with resolutions ranging from 10 m to 1000 m.

Rather than stacking millions of small image crops, we anchor our EO data sampling to the standard Sentinel-2 tile grid (110×110 km). This grounds the samples in a well-known geographic unit. To ensure a rich, diverse dataset not biased towards common land covers, we employed a stratified sampling strategy based on k-means clustering of land cover and RGB features. This actively over-samples

rare geographic and thematic classes. A total of 6273 globally distributed locations were sampled. For a sampled grid location and time, we acquired the corresponding Sentinel-2 data (Level 2A) along with overlapping Sentinel-1 SAR (GRD), Sentinel-3 OLCI (Level 1C) and SLSTR data. To ensure temporal consistency across modalities, we restrict the acquisition window for Sentinel-1 and Sentinel-3 imagery to the be within ± 1 days of the Sentinel-2 anchor timestamp for land areas and the same day for ocean areas. Sentinel-3 data is selected from a 25 times larger area to account for the coarser resolution. For each location, we also include the digital elevation model (DEM), and diverse land cover maps: WorldCover [33], GlobCover [1], MODIS [7], and ERA5-Land climate variables [20, 21].

To obtain temporal diversity, each location is sampled a random number of times, totaling 18332 tile and date combinations to support compute-adaptive pre-training and generalization across various climate and social applications. The total size of the dataset is approximately 22 TB. Data processing, spatio-temporal alignment, and sampling weights are detailed in the Suppl. Materials.

4. THOR foundation model

As established in the related work, THOR is the first architecture designed to simultaneously solve input heterogeneity and deployment versatility. The core novelty of THOR is the first successful extension, synthesis and scaling of three state-of-the-art concepts: 1) Per-band patch projection strategy inspired by USat to handle heterogeneous sensor data. 2) Extension of the flexible patching and weight-resizing strategy from FlexiViT to an MAE framework with random input image sizes, enabling dynamic input image sizes and patch sizes during inference. 3) GSD-aware 2D ALiBi encoding, inspired by CROMA, maintaining spatial context across GSDs and patch sizes. This section details the integration and the multi-pretext learning framework.

4.1. Encoder architecture and flexible patching

The core of THOR is a modified vision transformer (ViT) [6], built to solve both input heterogeneity (multi-sensor, multi-resolution) and deployment rigidity (fixed patch size and flexible input image size) simultaneously (Fig. 1).

4.1.1. Multi-sensor integration

Inspired by USat [12], THOR employs per-band patch projection layers to handle diverse sensor inputs. This flexibility allows the model to process any subset of bands during fine-tuning, accommodating computational constraints or missing data. The encoder supports grouping arbitrary sets of bands with the same GSD, allocating a larger number of patches for higher resolution bands to capture finer details, and fewer patches for coarser resolution bands. To reduce the resulting long token sequence length, an average

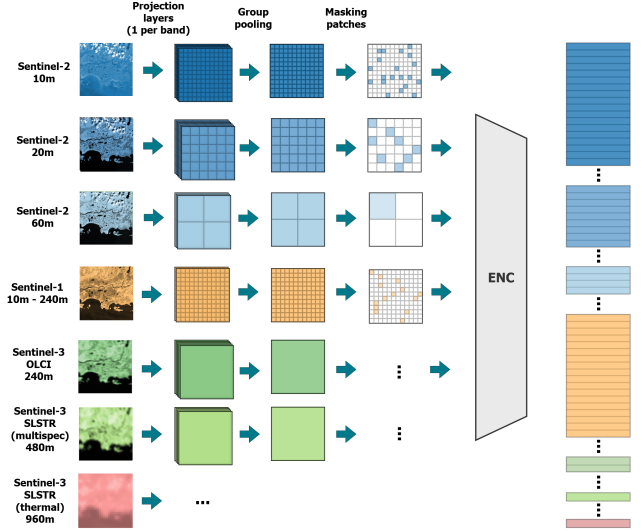


Figure 1. THOR encoder uses a single ViT. Data is processed using a band-wise patch projection layer and group average pooling. Patches are masked in a MAE fashion only during pre-training.

pooling step is applied to aggregate corresponding patches from the same band group (Fig. 1).

4.1.2. Compute-adaptive inference

To make THOR "compute-adaptive", allowing dynamic trade-offs between computational cost and accuracy without retraining, we incorporate the FlexiViT approach by randomizing the patch size during pre-training. The patch embedding weights are resized accordingly during training, enabling the resulting ViT to adapt to various patch sizes (e.g., from 4×4 to 32×32) at inference time using a single set of pre-trained weights. This flexibility has a crucial downstream benefit: a user can opt for a smaller patch size at inference time, producing a denser, higher-resolution token sequence, which can be processed by simpler decoders. We also randomize the input image size during pre-training, allowing THOR to extrapolate to larger input images than those used during fine-tuning.

4.1.3. GSD-aware positional encoding

USat's superpositional encodings assume fixed patch dimensions [12], and if you randomly change the patch size this scheme becomes impractical as it requires all patch sizes to be multiples of the smallest possible patch size. We therefore extend the 2D ALiBi (Attention by Linear Bias) approach by CROMA [8] to be GSD-aware.

Let a_{hij} denote the attention weight corresponding to query $\mathbf{q}_{hi} \in \mathbb{R}^d$ key $\mathbf{k}_{hj} \in \mathbb{R}^d$. The attention bias is calculated based on the real-world ground distance between patch centers as

$$a_{hij} = \mathbf{q}_{hi}^T \mathbf{k}_{hj} / \sqrt{d} - \frac{\text{dist}(\mathbf{x}_i, \mathbf{x}_j)}{\max(p)} \cdot m(h), \quad (1)$$

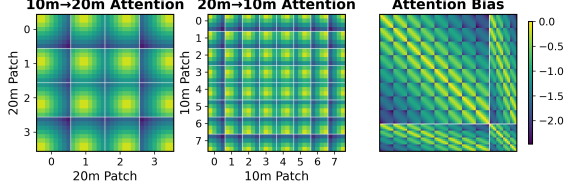


Figure 2. GSD-aware 2D-ALiBi for two groups: 10m GSD and 8x8 patches and 20m GSD and 4x4 patches. Left: Each sub-square is the ALiBi values (Eq. 1) between a 20 m GSD patch and all 10 m GSD patches. Mid: Each sub-square is the ALiBi values (Eq. 1) between a 10 m GSD patch and all 20 m GSD patches. Right: Full GSD-aware 2D-ALiBi matrix for the full sequence of $8 \times 8 + 4 \times 4 = 80$ patches, where the off-diagonal and diagonal blocks are the inter-product and intra-product attention biases, respectively.

where $\text{dist}(\mathbf{x}_i, \mathbf{x}_j)$ is the distance (in meters) between patches $\mathbf{x}_i$ and $\mathbf{x}_j$, $\max(p)$ is the largest patch size (in meters), and $m(h)$ denotes the strength of the positional biases to the h th self-attention head, called slopes m . We select slopes as in [8]. Fig. 2 validates this GSD-aware approach by comparing 2D-ALiBi values for two configurations of 10 m GSD with 8×8 patches and 20 m GSD with 4×4 patches, showcasing the relative positional encoding across tokens of different GSDs and patch dimensions.

For the lightweight decoder, which includes masked tokens for reconstruction, we modify the 2D sinusoidal positional encoding to be GSD-aware. Let g denote the GSD of the band we are reconstructing, and let pos denote the center position of a patch. The encoding v for that patch is:

$$\begin{aligned} v_x(pos, 2i) &= \sin \left(g \frac{pos + 0.5}{10000^{2i/D}} \right) \\ v_y(pos, 2i + 1) &= \cos \left(g \frac{pos + 0.5}{10000^{2i/D}} \right) \end{aligned} \quad (2)$$

The GSD-aware 2D-ALiBi not only elegantly solves the problem of handling flexible patch sizes and relating products of various resolutions, but also allows for test-time extrapolation to input sizes much larger than used during training [8]. As the decoder is discarded after pre-training, we opt for the simpler 2D sinusoidal positional encoding.

4.1.4. Randomized ground cover and patch size sampling

To accommodate the large span in GSD between modalities, we devise a data sampling strategy where we sample a random ground cover from the range $\sim (1000, 50000)\text{m}$. We then extract samples for the available modalities, making sure to only include data within a valid input image size between $\sim (20, 500)$ pixels.

We then select random patch sizes per modality GSD, making sure not to exceed a predefined token budget (see Suppl. Materials for the dynamic token budget sampling algorithm). The actual resizing of patch sizes is implemented

as a modified version of FlexiViT [4].

4.2. Decoder architecture

THOR is pre-trained using an extended MAE framework. This approach applies a self-supervised reconstruction objective combined with novel multi-modal prediction tasks.

Following the standard MAE framework [10], the decoder is substantially lighter than the encoder, focusing specifically on the reconstruction and land cover mapping task. This asymmetric design ensures that the high-quality feature representations reside solely within the heavier encoder, which is frozen for downstream tasks.

Unlike a standard MAE which uses a linear projection layer, our decoder head projects tokens back to the patch space using a Conv2D-Transpose layer.

4.3. Loss formulation

THOR is trained from multiple pretext tasks to ensure generality and robustness across different applications (Fig. 3):

- Pixel-level reconstruction: Pixel-level input band reconstruction is performed using our proposed flexible ViT MAE loss (Sec. 4.3.1). This task forces the model to learn fine-grained details for pixel-level applications.
- Patch-level contrastive learning: We devise a patch level guided soft (multi label) contrastive loss to leverage rich semantic information in the available land cover products (Sec. 4.3.2).
- Pixel-level map prediction: The model predicts land cover maps, such as ESA WorldCover. This provides dense, geo-semantic supervision across different GSDs (e.g., WorldCover 10m maps predicted from Sentinel-1/2 groups, and MOD12Q1 maps predicted from Sentinel-3 SLSTR). The model also predicts elevation and slope derived from a DEM at 10 m and 60m GSD (Sec. 4.3.3).
- Image-level prediction [22, 27]: Predicts ERA5-Land variables (from daily statistics, e.g., soil water, temperature, precipitation), latitude, longitude, and month, providing coarse-grained, climate-relevant feature learning.
- Image level SAR task: The learning includes Sentinel-1 SAR ascending/descending orbit direction classification task and incidence angle prediction task.

Similar to USat [12] we use group specific projection layers to map decoder tokens from group g to patches of shape (P_g, P_g, C_g) where C_g is the number of bands in this group. The input bands are then patchified to the same patch sizes and a channel-wise MSE loss over all the masked patches is used.

We add linear heads to the pooled encoder tokens for image-level tasks, using cyclic encoding for angular/temporal variables [22]

$$\text{cyclic_encoding}(\mathbf{x}, s) = \begin{bmatrix} \sin \left(\frac{2\pi}{s} \mathbf{x} \right) \\ \cos \left(\frac{2\pi}{s} \mathbf{x} \right) \end{bmatrix}, \quad (3)$$

where the factors $\mathbf{x} \in \mathbb{R}^{B \times C}$ and $s \in \mathbb{R}^+$ serve as scaling parameters, with the output being of dimension $\in \mathbb{R}^{B \times 2C}$.

4.3.1. Flexible ViT MAE loss

FlexiViT encoders present a challenge for MAE pretraining: the decoder must reconstruct patches of varying spatial sizes. While arbitrary patchification easily matches the number of decoder tokens to target patches, the standard linear projection heads produce outputs of fixed size. To address this, we replace the linear head with a transposed Conv2D layer, which allows bilinear interpolation of projection weights. A FlexiViT-inspired rescaling then keeps the loss approximately consistent across patch sizes.

Standard MAE uses a pixel-wise MSE loss to reconstruct masked patches: $\mathcal{L}_{mae} = \frac{1}{N} \|\text{vec}(\mathbf{x}) - \langle \mathbf{v}, \mathbf{z} \rangle\|^2$, where $\mathbf{x} \in \mathbb{R}^{p \times p}$ is the target patch, $\mathbf{z} \in \mathbb{R}^{D_d}$ is the decoder token embedding, and $\mathbf{v} \in \mathbb{R}^{D_d \times p^2}$ maps decoder embeddings to pixel predictions. When the patch size p changes to a new size p^* , the prediction weights $\mathbf{v}$ must also change to $\mathbf{v}^*$ to maintain the correct projection. We use a bilinear interpolation to change the weights, such that $\mathbf{v}^* = \mathbf{v}\mathbf{B}^T$, where $\mathbf{B} \in \mathbb{R}^{p^2 \times p^2}$. We observe that by scaling the entire loss term with the pseudo-inverse $\mathbf{B}^+$, the MSE loss for the resized patch is equivalent to the loss for the original patch when $\mathbf{B}$ has full column rank (i.e., $p^* \geq p$, and a sound approximation for $p^* < p$:

$$\begin{aligned} \mathcal{L}_{mae}^* &= \frac{1}{N} \|\mathbf{B}^+(\mathbf{B}\text{vec}(\mathbf{x}) - \langle \mathbf{v}\mathbf{B}^T, \mathbf{z} \rangle)\|^2 \\ &= \frac{1}{N} \|\text{vec}(\mathbf{x}) - \langle \mathbf{v}, \mathbf{z} \rangle\|^2 = \mathcal{L}_{mae}. \end{aligned} \quad (4)$$

This formulation ensures a stable, approximately equal reconstruction objective across randomized patch sizes.

4.3.2. Patch-level contrastive loss

Inspired by Galileo, we use a patch-wise contrastive loss to enhance local detail and improve token discrimination within each sample. This improves the modeling of fine-grained features [27]. We further extend the multi-label guided approach in [29] to the patch level. We randomly divide the unmasked patch tokens into K partitions and compute an average embedding for each partition. Concurrently, we extract an average land cover histogram from the corresponding patch locations for each of the K partitions, and soft similarity labels are then generated using cosine similarity between these K normalized histograms. A contrastive loss (Eq. 5) is applied to the K average embeddings, forcing the model to produce similar representations for partitions with similar land cover compositions. This loss is computed per band group, for all land cover tasks, adhering to the same viability constraints as our map prediction task.

The contrastive loss for group g and task t is

$$\mathcal{L}_{con}^{g,t} = -\frac{1}{M} \sum_{i=1}^M \log \frac{\sum_{j \in P(i)} \exp(-h_{i,j}^{g,t} f(\mathbf{x}_j, \mathbf{x}_i)/\tau)}{\sum_{k \in Q(i)} \exp(-f(\mathbf{x}_k, \mathbf{x}_i)/\tau)}, \quad (5)$$

where $M = B_{g,t} \times K$ is the total batch size times the number of partitions, $P(i) = \{j : 1 \leq j \leq M, j \neq i, y_j = y_i\}$, $Q(i) = \{k : 1 \leq k \leq M, k \neq i, y_k \neq y_i\}$, $f(\cdot)$ denotes the cosine similarity, $h_{i,j}^{g,t}$ is the $[0, 1]$ normalized soft similarity label between the averaged set of patches i and j , and y_l is equal to the local GPU device. I.e., we only select positive pairs from the same local device, and select negatives from all other devices, as all samples in the local batch stem from the same tile and date, whereas batches from other devices will be from other tiles and dates. This ensures that we do not enforce the features to be season and geographically invariant.

The total contrastive loss is thus

$$\mathcal{L}_{con} = \frac{1}{G} \sum_{g=1}^G \frac{1}{|T_g|} \sum_{t \in T_g} \mathcal{L}_{con,g,t} \quad (6)$$

where G is the total number of band groups, T_g is the set of viable land cover tasks for the corresponding group

4.3.3. Map prediction loss

We incorporate pixel-level pretext tasks [13, 22, 27], providing dense, semantic supervision via multiple land cover products (e.g., WorldCover, GlobCover), and DEM-based elevation and slope regression.

These are treated as additional targets, and the decoder uses separate projection heads to predict each map, adapting the FlexiViT strategy to project variable-GSD decoder embeddings to fixed target GSDs. A compatibility constraint permits only projections resulting in a target patch size within a $[4, 32]$ pixel range. For example, predicting a 10 m WorldCover map from a 16-pixel, 60 m GSD input patch is disallowed, as it would require an unstable 96×96 pixel projection.

Unlike the MAE reconstruction on masked tokens, the map prediction loss covers encoder patches (masked and unmasked). We apply a CE loss (0.1 label smoothing) for classification tasks and MSE loss for regression tasks, standardizing DEM targets (elevation and slope) using dataset statistics following per-sample min-normalization of the elevation (see Suppl. Materials).

4.3.4. Total loss

Our final pre-training loss, $\mathcal{L}_{total}$, is a weighted sum of the MAE reconstruction loss ($\mathcal{L}_{mae}$), the map prediction losses ($\mathcal{L}_{map,t}$), the ERA5 land, month, coordinate and S1 incidence regression loss ($\mathcal{L}_{era5}, \mathcal{L}_m, \mathcal{L}_{coord}, \mathcal{L}_{inc}$), S1 orbit direction loss ($\mathcal{L}_{orb}$), and our novel contrastive loss ($\mathcal{L}_{con}$). $\mathcal{L}_{total} = \lambda_1 \mathcal{L}_{mae} + \lambda_2 \mathcal{L}_{con} + \sum_t \lambda_{3,t} \mathcal{L}_{map,t} + \lambda_4 \mathcal{L}_{era5} +$

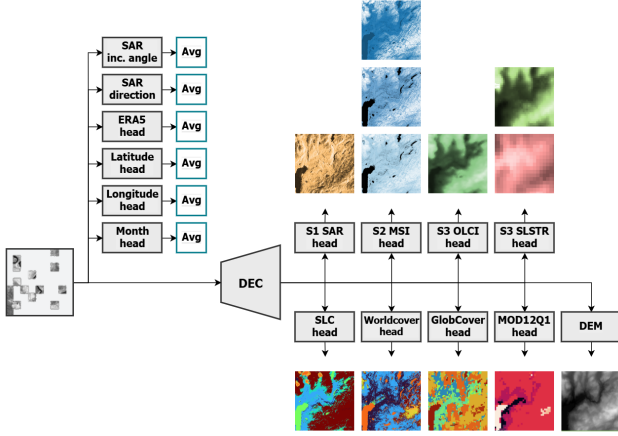


Figure 3. Pretext tasks used for learning THOR.

$\lambda_5 \mathcal{L}_m + \lambda_6 \mathcal{L}_{coord} + \lambda_7 \mathcal{L}_{inc} + \lambda_8 \mathcal{L}_{orb} + \lambda_9 \mathcal{L}_{fft}$, where $t \in \{WC, GC, MCD, DEM, SCL\}$ are the land cover prediction tasks. $\mathcal{L}_{fft}$ is a L1 MAE reconstruction loss in the Fourier domain, included for stability during training [14]. The weights (λ_k) are provided in Suppl. Materials.

5. Experiments

Our experiments are designed to validate THOR’s core hypotheses: **1)** Is THOR more data-efficient in low-label regimes, validating our "data-hungry decoder" hypothesis? **2)** Does our architecture successfully synthesize and learn from the full 10 m - 1000 m Sentinel-1, -2, and -3 sensor suite?, and **3)** Is the compute-adaptive mechanism the key driver of this performance?

5.1. Experimental setup

We evaluate on two main benchmarks: PANGAEA [18] and Copernicus-Bench [30]. PANGAEA consists of a diverse suite of 9 semantic segmentation tasks, including HLS, MADOS, PASTIS, and Sen1Floods11. We follow its standard protocol, evaluating on the 10%, 50%, and 100% labeled data splits to test our data-efficiency hypothesis. To specifically validate THOR’s ability to process Sentinel-3 data, we evaluate on the four Sentinel-3 OLCI-specific tasks (Cloud-S3, LC100Cls-S3, LC100Seg-S3, Biomass-S3) from the Copernicus-FM benchmark.

We compare THOR against a wide range of state-of-the-art (SOTA) FMs, including Copernicus-FM [30], DOFA [32], and TerraMind [13], using the scores reported in [18] and [30].

5.2. Implementation details

We pre-train a family of models, THOR-Tiny, -Small, -Base, and -Large (THOR-T/S/B/L) from scratch on the THOR Pretrain dataset for 400 epochs. All models are

trained using the AdamW optimizer with a base learning rate of $3e-4$ for the Base and Large model, and $4e-4$ for Small and Tiny, a weight decay of 0.05, and a linear warmup of 40, 20, 10, 10 epochs for THOR-Large, -Base, -Small and -Tiny models, respectively followed by a cosine decay schedule.

Training was conducted on 16 AMD MI250X GPUs, with a total batch size of 1024. During pre-training, we randomly sample both input resolution ($32 \times 32 - 1024 \times 1024$) and patch size ($4 \times 4 - 32 \times 32$) for every sample. To manage computational constraints introduced by randomized patch sizes and input image sizes across multiple band groups, we implemented a simple token budget heuristic. During training, a threshold for the maximum number of tokens is enforced. When sampling a product group, a patch size is drawn such that the resulting number of tokens does not exceed the remaining budget, ensuring efficient use of memory across heterogeneous inputs.

5.3. Main result: State-of-the-art in data-limited scenarios

Our central hypothesis is that THOR’s flexible patching overcomes the "data-hungry decoder" problem faced by rigid models. We test this on the 10% Pangaea benchmark split using a 6×6 patch size.

In this low training data regime, THOR-B achieves the best average rank across all datasets (Tab. 1), outperforming all published models, including TerraMind. This performance, particularly on fine-grained tasks like Sen1Floods11 (86.29 mIoU), validates that THOR is significantly more data-efficient than its fixed-patch counterparts.

5.4. Full-data benchmarking

We next evaluate THOR’s performance in Pangaea using full training data availability and for the Sentinel-3 OLCI scenarios from Copernicus-Bench [30].

For 100% training data, the THOR family remains state-of-the-art or highly competitive. THOR-B achieves the top rank on PASTIS (40.76% mIoU) and THOR-L on CropMap (60.75% mIoU), demonstrating that its architecture scales effectively with more data. However, the performance of all models varies across all tasks, and the variation of THOR is similar to other FMs, in both high- and low-data regimes. A full comparison against all baseline models is provided in the Suppl. Materials.

The Copernicus-Bench validates THOR’s ability to synthesize Sentinel-3 OLCI data (Tab. 2). Our model outperforms all baselines on two of the four OLCI specific tasks, including a +4.1 mIoU gain on Cloud-S3 and a -6.6 RMSE (improvement) on Biomass-S3 over the Copernicus-FM baseline. This confirms that THOR effectively learns from the challenging 10 m - 1000 m GSD range.

To further evaluate the learned representations, we

Table 1. Pangaea results with 10% training data in mIoU. Bold/underline mark best/second-best per column.

Model	HLS Burns	MADOS	PASTIS	Sen1Floods11	FBP	DynEarthNet	CropMap	SN7	AI4Farms	Avg. Rank
CROMA	76.44	32.44	32.80	<u>87.22</u>	37.39	36.08	36.77	42.15	38.48	6.11
DOFA	71.98	23.77	27.68	82.84	27.82	39.15	29.91	46.10	27.74	10.22
GFM-Swin	67.23	28.19	21.47	62.57	55.58	28.16	27.21	39.48	32.88	12.56
Prithvi	77.73	21.24	33.56	86.28	29.98	32.28	27.71	36.78	35.04	10.22
RemoteCLIP	69.40	20.57	17.19	62.22	<u>56.23</u>	34.43	19.86	43.11	23.85	12.33
SatlasNet	74.79	29.87	16.76	83.92	37.86	34.64	29.08	49.78	13.91	10.22
Scale-MAE	75.47	21.47	22.86	64.74	48.75	35.27	13.44	49.68	26.66	10.78
SpectralGPT	83.35	20.29	34.53	83.12	39.51	35.33	31.06	36.31	37.35	8.56
S12-MoCo	73.11	19.47	32.51	79.58	35.57	32.24	36.54	49.46	37.97	10.67
S12-DINO	75.93	23.47	36.62	84.95	34.63	32.78	<u>38.44</u>	41.15	37.91	8.33
S12-MAE	76.60	18.44	31.06	84.81	35.56	30.59	35.29	40.51	23.60	11.44
S12-Data2Vec	74.38	17.86	33.09	81.91	37.27	33.63	34.11	40.66	22.85	12.11
TerraMind-B	77.39	44.06	39.96	84.43	54.00	<u>37.35</u>	35.65	43.21	38.59	<u>4.00</u>
UNet Baseline	<u>79.46</u>	24.30	29.53	88.55	52.58	35.59	13.88	46.08	34.84	7.11
ViT Baseline	75.92	10.18	38.44	81.85	56.53	35.39	27.76	36.01	39.20	8.78
THOR-B	76.90	40.67	<u>38.93</u>	86.29	42.80	35.21	42.23	<u>55.94</u>	<u>38.90</u>	3.78
THOR-T	75.98	<u>41.65</u>	36.26	82.70	42.81	34.03	37.82	58.52	38.56	5.78

Table 2. Benchmark results on selected Copernicus-Bench benchmarks. †: We use a patch size of 10 for Cloud-S3, 8 for LC100Cls-S3, 6 for LC100Seg-S3 and 4 for Biomass-S3.

	Metric	Supervised	Supervised	Random	DOFA	Copernicus	THOR
Backbone	—	ViT-S/16	ViT-B/16	ViT-B/16	ViT-B/16	ViT-B/8	ViT-B/10†
Modality	—	—	—	—	All (spectral)	All	All (spectral)
Cloud-S3	mIoU	61.7 ± 0.7	63.0 ± 0.8	60.9 ± 0.0	58.2 ± 0.1	62.0 ± 0.7	66.1 ± 0.1
LC100Cls-S3	mAP	91.3 ± 0.3	91.4 ± 0.5	88.9 ± 0.1	89.5 ± 0.0	93.3 ± 0.4	91.0 ± 0.0
LC100Seg-S3	mIoU	20.1 ± 0.4	19.3 ± 0.5	18.2 ± 0.1	16.5 ± 0.1	24.1 ± 0.0	19.9 ± 0.0
Biomass-S3	RMSE ↓	68.1 ± 0.3	68.3 ± 0.4	68.7 ± 0.5	74.1 ± 0.1	66.3 ± 0.1	59.7 ± 0.0

benchmark THOR on the GEO-Bench image classification datasets using a k -Nearest Neighbor (k NN) classifier [15].

- THOR-S achieves best accuracy scores on m-Brick-Kiln. For only 1% training data, THOR-S achieves 91.5%, significantly outperforming other foundation models like DOFA-L (80.6%) and Galileo-N (79.4%).
- On the m-EuroSat 1% split, THOR-T (52.8%) achieves second best accuracy score, only outperformed by Galileo-B (56.6%).

Details are provided in the Suppl. Materials.

5.5. Ablation studies

Architecture vs. dataset scale To isolate architectural gains from pre-training dataset scale, we decouple compute-adaptivity from volume. Fig. 4 demonstrates that simply reducing the inference patch size from 16×16 to 4×4 , using a single set of frozen weights, yields a +4 mIoU points gain. This proves that dense tokenization is a primary performance driver, independent of data volume.

Value of multi-sensor synthesis. A core design principle of THOR is its ability to ingest and synthesize synergistic information from heterogeneous sensors. To validate this capability, we conducted an ablation study on the Sen1Floods11 benchmark using the 10% data split. We fine-tuned the THOR-B model with a UPerNet decoder on three different input modality configurations: S1 only, S2 only, and the combined S1 + S2 inputs. The combined

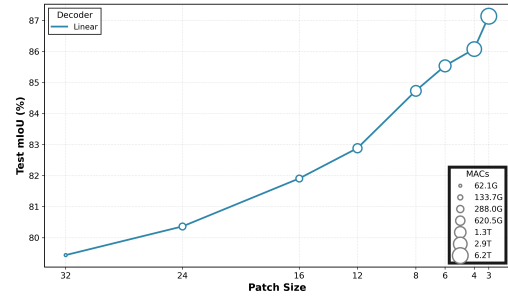


Figure 4. Test mIoU for THOR-B with varying patch sizes using a fixed number (18) of tokens with linear probing segmentation on the Sen1Floods11 dataset (S1 and S2), 10% of the training data, with mean aggregation of features. Sliding window inference MACs indicated by circle sizes.

S1 + S2 configuration (Tab. 3) outperforms single-sensor baselines, proving the encoder learns complementary features across radar and optical streams.

Table 3. Sen1Floods11 10% test mIoU results for different modality configurations using THOR-B model with UPerNet decoder.

Modalities	mIoU
Sentinel-1	78.09
Sentinel-2	<u>87.25</u>
Sentinel-1 + Sentinel-2	87.70

Supplementary experiments on PANGAEA further validate THOR: concatenation outperforms mean aggregation, THOR-B is more effective in low-data regimes (10%) while THOR-L performs best with full training data (100%), and a simple linear probe often achieves competitive results with $\sim 1000\times$ fewer parameters than UPerNet (see Suppl. Materials).

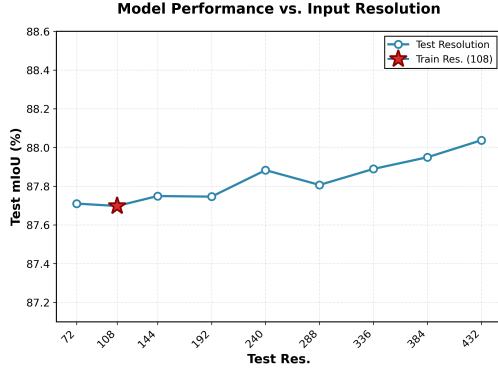


Figure 5. Test mIoU results of a THOR-B (w/UPerNet), trained on a 108×108 image size, evaluated on increasingly larger images.

5.6. Test-time extrapolation to larger images

We validated our GSD-aware 2D-ALiBi’s ability to extrapolate to larger input sizes than seen during training, a key property of relative positional encodings [8]. We fine-tuned a UPerNet decoder on the Sen1Floods11 (10% split) dataset using a frozen THOR-B encoder with S1 + S2 inputs. The model was trained only on 108×108 pixel crops with a patch size of 6.

During evaluation, we tested this fixed model on the test set using various input sizes, applying Pangaea’s sliding window inference up to the full 512×512 image size. As shown in Fig 5, performance does not degrade at larger scales; on the contrary, it shows a consistent improvement as the input size increases, confirming that our GSD-aware 2D-ALiBi maintains spatial coherence at scales far beyond the training distribution.

5.7. Use case: mapping of snow cover

We validate THOR’s compute-adaptive capability on a data-scarce climate task: snow cover fraction regression using 500 m / 1000 m GSD Sentinel-3 SLSTR data.

We fine-tune THOR-B, comparing a linear decoder against UPerNet [31]. The results in Tab. 4 provide two key insights:

- Reducing inference patch size from 16×16 to 4×4 with the UPerNet decoder lowered RMSE from 12.4 to 9.90. This reduction confirms that denser token sequences benefit even coarse-resolution data.
- A linear decoder with 4×4 patches performs comparably to the larger UPerNet with 4×4 patches (9.88 vs 9.90 RMSE). This confirms our hypothesis: complex decoders often compensate for "token-starved" encoders. By providing a dense token sequence, THOR’s flexible patching enables simpler decoders, validating its efficiency.
- Furthermore, THOR outperforms both the UNet and DOFA baselines, which yield significantly higher errors of 12.4 and 13.8 RMSE, respectively.

Table 4. RMSE snow cover fraction. Image size 128×128 and concatenated the tokens of the 500 m and 1000 m bands.

Decoder	Patch size	RMSE
UNet		12.4
DOFA	16×16	13.8
UPerNet	16×16	14.0
	8×8	12.4
	4×4	9.90
Linear decoder	4×4	9.88

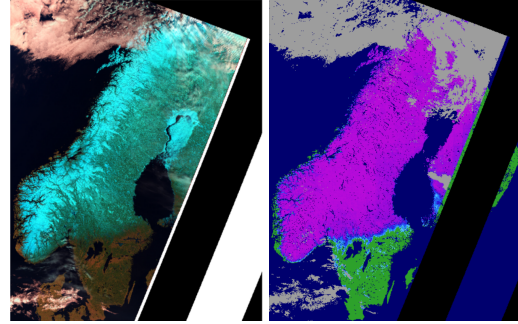


Figure 6. Left: False color SLSTR image. Right: Predicted snow cover fraction, where green is less than 15% and purple is 100%.

6. Conclusion

In this work, we addressed a weakness of current EO foundation models: their architectural rigidity. We argued that fixed patch sizes lead to data-hungry decoders, limiting their utility in data-scarce scenarios. We proposed THOR, the first FM to synthesize a compute-adaptive patching strategy with a multi-sensor architecture that unifies Sentinel-1, -2, and -3 (OLCI & SLSTR) data.

Our experiments validate our central hypothesis. THOR achieves state-of-the-art performance in the Pangaea 10% benchmark, demonstrating superior data efficiency. This confirms that the ability to use smaller patch sizes at inference provides a denser token sequence that is more effective for fine-tuning on limited data. We also proved our complex multi-sensor synthesis was successful, with THOR outperforming baselines on two of four Sentinel-3 OLCI specific tasks, validating its unique 10 m - 1000 m GSD capability.

THOR, while versatile, has limitations that open clear avenues for future work. While our dataset includes temporal samples, the architecture itself does not explicitly model time. Future work will focus on extending this flexible-patching concept to an explicit spatio-temporal backbone and integrating other key modalities like Sentinel-5P (air quality) or passive microwave data (climate) to further strengthen THOR applicability to climate and society challenges.

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